Óscar Fraile

Full Stack Engineer

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Profile

I'm a software engineer who enjoys working on the backend, figuring out how the pieces fit together, and turning ideas into working products. I also care about thoughtful design and making those products intuitive to use. Lately, I've been building tools with AI.

Professional experience

Full Stack Engineer | Educaria

2019 - Present · Madrid, Spain

Building Alexia - the leading EdTech platform for schools and universities in Spain and Latin America, with 2M+ active users across web, Android, and iOS.

- Contributing to the transition from a .NET Framework 4.8 monolith to distributed microservices with .NET Core and Entity Framework Core, applying Hexagonal Architecture, Clean Architecture, SOLID, DDD and CQRS.
- Full stack development with React and TypeScript, authentication with Identity Server and caching with Redis.
- CI/CD pipelines in Azure DevOps, Docker containers and deployments to Kubernetes managed with Rancher.
- Unit tests with xUnit, integration tests, and API and load-testing scenarios with k6.
- AI-assisted development with Claude Code, MCP and shared skills, using Spec-Driven Development to guide implementation and review with clear requirements and acceptance criteria.
- Development of a RAG chatbot with Python and FastAPI for teachers and students, supported by knowledge of chunking, embeddings, metadata, reranking and grounding, as well as agent orchestration with LangChain and LangGraph, query rewriting and context management to balance response quality, latency and cost.

Solutions Assistant | NTT Data

2017 - 2019 · Madrid, Spain

Software developer on a large-scale project for Repsol, powering point-of-sale across gas stations in Spain and Peru.

- Built mission-critical point-of-sale, handheld, and back-office systems.
- Joined the early phase of the new back-office front-end, built with Angular.
- Handled development, unit testing, and documentation within agile teams.

Technical skills

Frontend: React, TypeScript, Angular, Ionic, JavaScript, Tailwind CSS

Backend: C#, .NET, Entity Framework Core, Python, FastAPI, SQLAlchemy, Identity Server, Node.js, OData, GraphQL

Architecture & DevOps: Microservices, Hexagonal Architecture, Clean Architecture, SOLID, Docker, Kubernetes, Rancher, CI/CD, Azure DevOps, GitHub Actions, Azure, Google Cloud, Vercel

Databases: SQL Server, PostgreSQL, Redis, Firestore, MongoDB

AI & Automation: Claude Code, MCP, Spec-Driven Development, RAG, LangChain, LangGraph, GitHub Copilot, Prompt Engineering, AI Agents

Testing & Quality: xUnit, Moq, k6, BDD / Gherkin, SonarQube

Selected projects

GoIA

Speaker diarization platform - from the model to the API

Async, job-based Batch API in FastAPI on Cloud Run: create job → upload audio → run → poll/webhook → results.

On-demand GPU orchestration: spins up ephemeral Compute Engine VMs (T4→L4 zone cascade) that run the model in Docker and self-delete when done.

2026 World Cup Predictor

Match predictions for the 2026 World Cup, with an honest track record

Two-phase Poisson GLM: a pre-trained prior from international matches, blended with in-tournament learning via shrinkage as more games are played.

Dixon-Coles correction for low-scoring matches and a full scoreline matrix to derive 1X2 probabilities and favorites.

Education

Bachelor's Degree in Computer Engineering | 2026

UDIMA - Universidad a Distancia de Madrid

Higher Diploma in Multiplatform Application Development | 2017

Centro Europeo de Estudios Profesionales

Certifications

Microsoft Certified: Azure Developer Associate · Microsoft · 2026

Claude Code in Action · Anthropic · 2026

Machine Learning with Python · IBM · 2026

Mathematics for Understanding the World: Models, Data, and Artificial Intelligence · UDIMA - Universidad a Distancia de Madrid · 2026

What The Hack: Azure OpenAI Fundamentals · Código Facilito · 2024

Cambridge English: Advanced (CAE) - CEFR C1 · University of Cambridge · 2018

Languages: Native Spanish · English C1 (Cambridge CAE)